

Numerical PDEs Homework #0

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Problem 1: Finite Differences

Write a MATLAB code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

1. Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.

Solution: The finite difference approximation closely follows the exact analytic derivative on the interval $0 \leq x \leq 2\pi$ as seen in Figure 1. Looking closely, we can see larger errors at the endpoints because we used first-order one-sided differences instead of second-order centered differences.

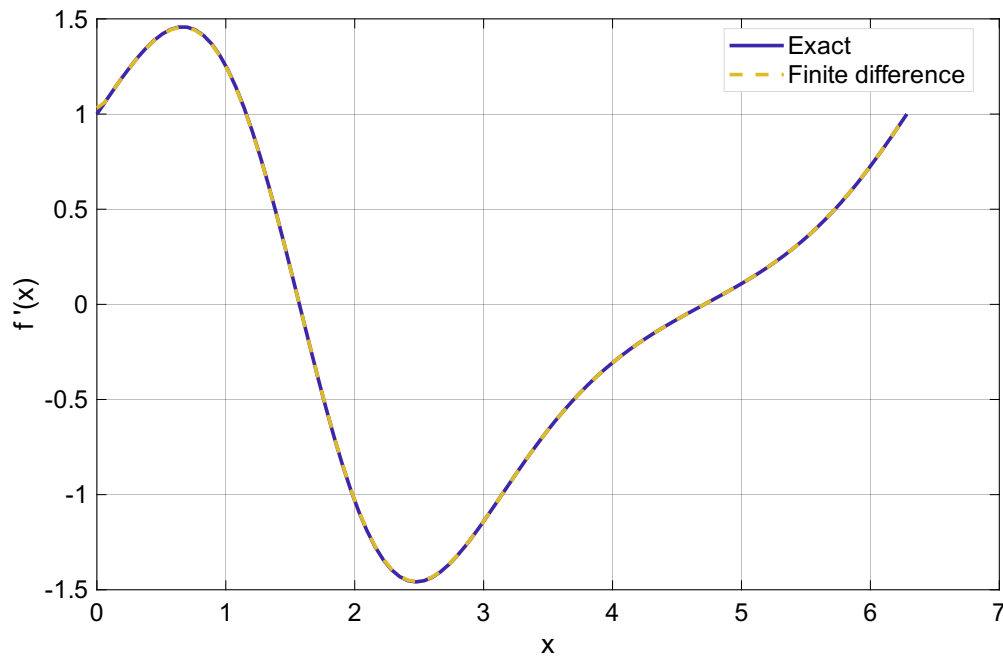


Figure 1: Exact analytic solution vs. numerical solution

2. Show a relative error convergence plot by plotting the relative error on the y -axis and the grid size N on the x -axis. Measure relative error as

$$\text{Err} = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

Plot the error using both the $p = \infty$ and $p = 2$ norms on a log-log scale.

The relative errors should be lines with slopes 1 and $3/2$, respectively; i.e., it scales with $\frac{1}{N}$ and $\frac{1}{N^{3/2}}$.

Solution: The relative errors decrease as the granularity of the grid increases. The infinity-norm slope is approximately -1.011 , showing first-order convergence, while the 2-norm slope is approximately -1.580 , which is close to the $-3/2$ convergence rate.

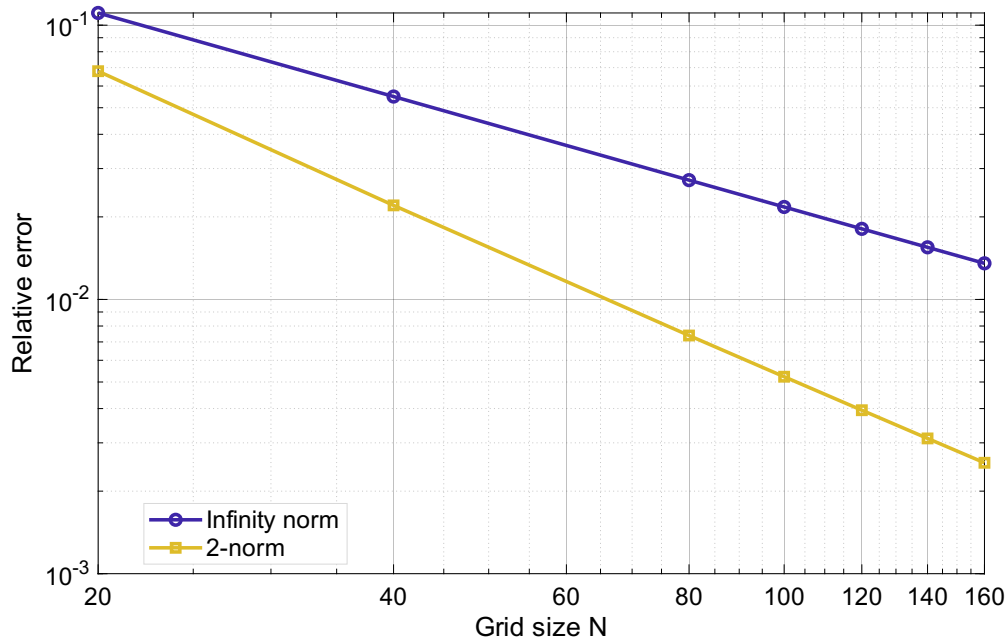


Figure 2: Relative error convergence plot

Extra Credit: The $3/2$ slope is strange. Why is that happening?

Solution:

The relative 2-norm error is

$$\text{err}_2 = \frac{\|f' - f'_n\|_2}{\|f'\|_2}$$

$f'(x_i) \sim O(1)$ and $f'_n(x_i) \sim O(1)$ at each of the $N + 1$ grid points because their values are not affected by grid spacing.

For the denominator, we can write:

$$\begin{aligned}\|f'\|_2 &= \sqrt{\sum_{i=0}^N f'(x_i)^2} \\ &\sim \sqrt{(N+1) \cdot O(1)} \\ &\sim O(N^{1/2})\end{aligned}$$

For the numerator, we can write:

$$\|f' - f'_n\|_2 = \sqrt{\sum_{i=0}^N (f'(x_i) - f'_n(x_i))^2}$$

We know that the second order approximations at the $N - 1$ interior points have error $O(h^2) = O(N^{-2})$ and the first order boundary approximations at both ends have error $O(h) = O(N^{-1})$. So, we can say:

$$\begin{aligned}\|f' - f'_n\|_2 &\sim \sqrt{(N-1)O(N^{-2})^2 + 2O(N^{-1})^2} \\ &\sim \sqrt{O(N^{-3}) + O(N^{-2})} \\ &\sim O(N^{-1})\end{aligned}$$

Plugging in for the numerator and denominator gives:

$$\text{err}_2 \sim \frac{O(N^{-1})}{O(N^{1/2})} \sim \boxed{O(N^{-3/2})}$$

Since $h = O(N^{-1})$, we can rewrite to say

$$\boxed{\text{err}_2 = O(h^{3/2})}$$

Problem 2: Second-Order Derivatives

Repeat Problem 1, but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Solution: The finite difference approximation closely follows the exact analytic second derivative on the interval $0 \leq x \leq 2\pi$ as seen in Figure 3. Assuming periodicity allows us to use centered second-order differences at every grid point.

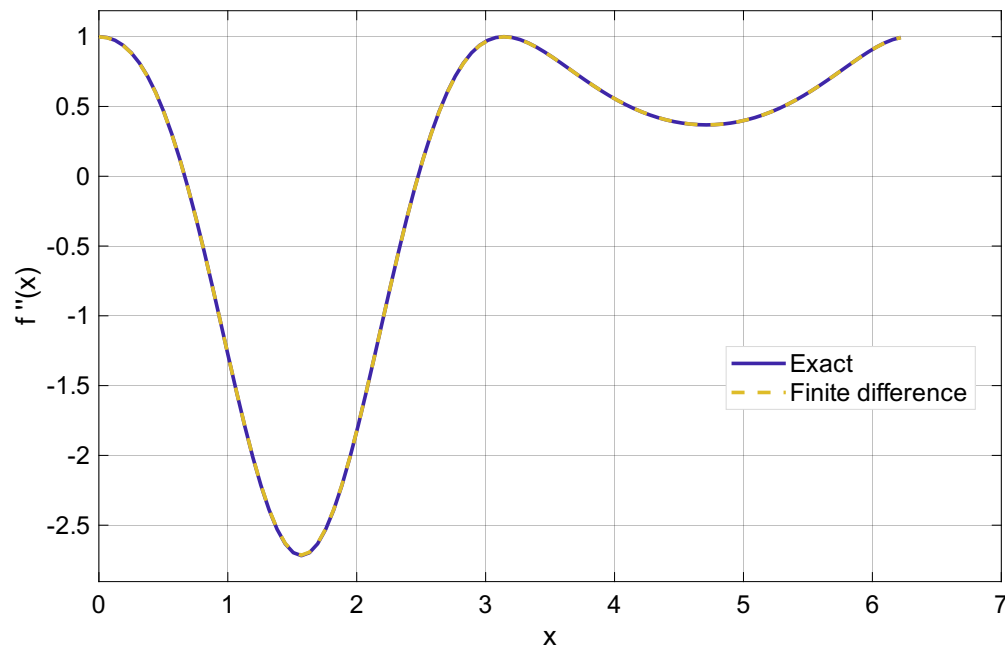


Figure 3: Exact analytic solution vs. numerical solution

Relative errors decrease as the grid granularity increases. The infinity-norm slope is approximately -1.995 and the 2-norm slope is approximately -1.996 , which are both close to the expected slope of -2 . Therefore, the finite difference approximation demonstrates second-order convergence. The error is $O(N^{-2})$ for both norms because we assumed periodicity and used the second-order finite difference method at every point.

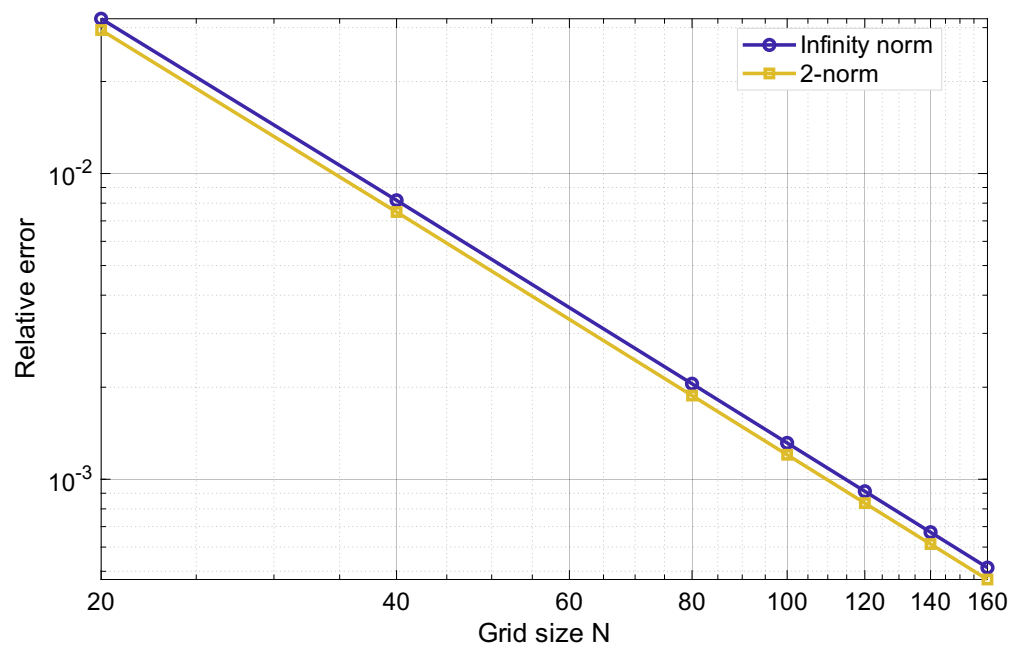


Figure 4: Relative error convergence plot

Problem 3: ODE

Consider the ODE

$$\frac{d^2u}{dx^2} + \sin(x)\frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding the boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Solution: Using the test solution $u(x) = \sin(x)$, we have $u'(x) = \cos(x)$, and $u''(x) = -\sin(x)$. Substituting into $u'' + \sin(x)u' + u = f(x)$ gives

$$-\sin(x) + \sin(x)\cos(x) + \sin(x) = f(x).$$

Therefore, $f(x) = \sin(x)\cos(x)$. The Dirichlet boundary conditions are $u(0) = \sin(0) = 0$ and $u(5) = \sin(5)$.

The finite difference solution closely matches the exact solution $u(x) = \sin(x)$ on the interval on $0 \leq x \leq 5$ as seen in Figure 5.

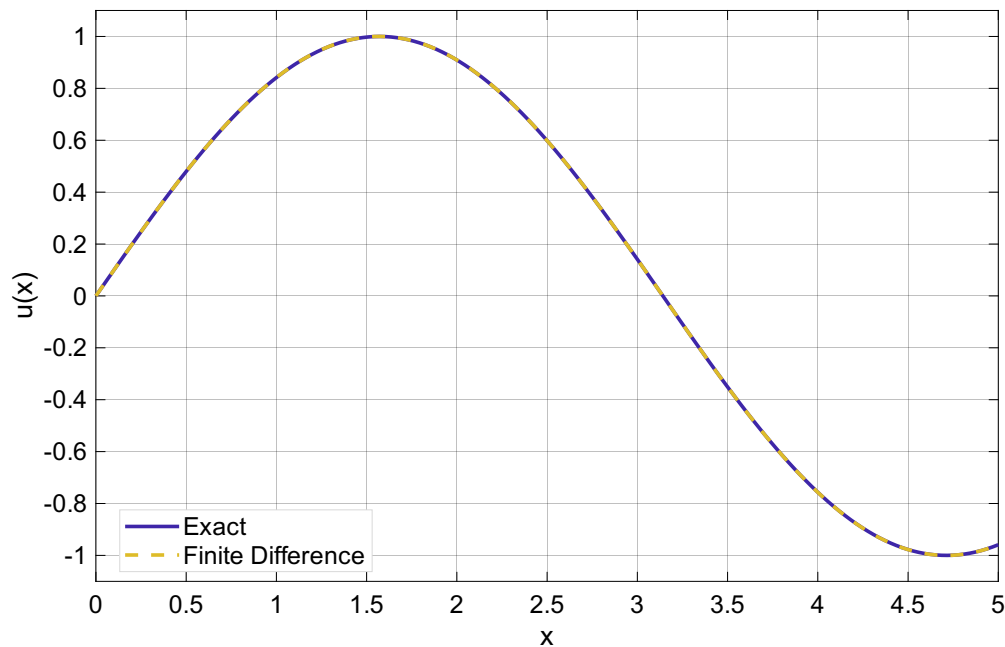


Figure 5: Exact analytic solution vs. numerical solution

The infinity-norm slope is approximately -2.045 and the 2-norm slope is approximately -2.041 , which are both close to the expected slope of -2 . Therefore, the finite difference approximation demonstrates second-order convergence.

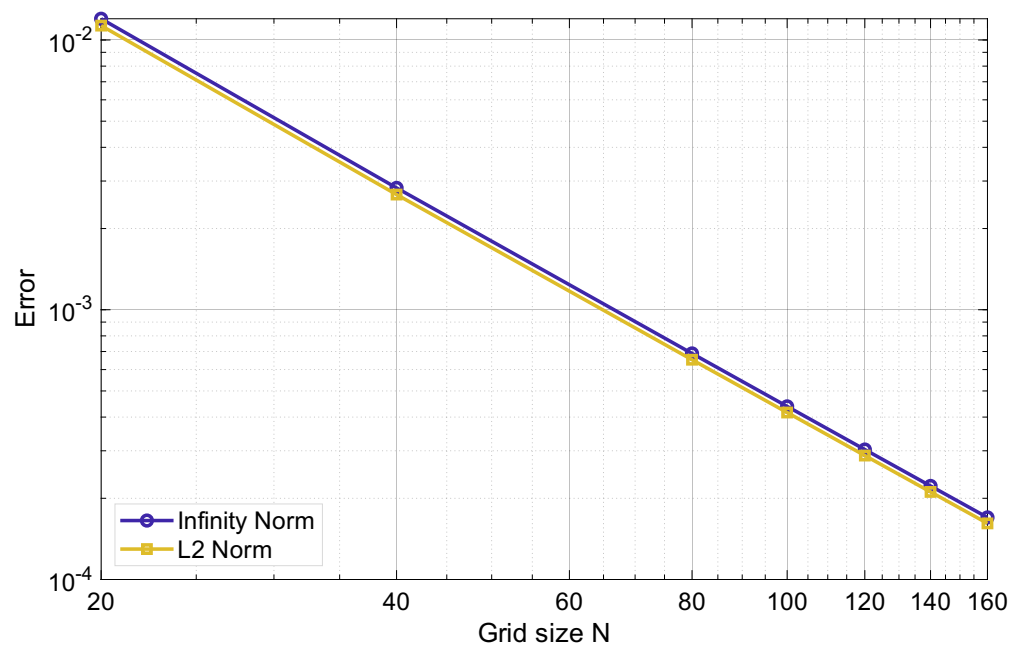


Figure 6: Relative error convergence plot

Problem 4: Poisson Equation

Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5]$, $y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.

1. Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second-order convergence plot similar to the first three problems.

Solution: Given that the test solution is $u(x, y) = \sin(x) \cos(y)$, the second derivatives are

$$u_{xx} = -\sin(x) \cos(y)$$

and

$$u_{yy} = -\sin(x) \cos(y).$$

Thus,

$$\nabla^2 u = u_{xx} + u_{yy} = -2 \sin(x) \cos(y) = f(x, y).$$

The Dirichlet boundary conditions are:

$$u(0, y) = 0$$

$$u(5, y) = \sin(5) \cos(y)$$

$$u(x, 0) = \sin(x)$$

$$u(x, 5) = \sin(x) \cos(5)$$

We use the centered second-order finite difference method for interior points:

$$u_{xx}(x_i, y_j) \approx \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2},$$

$$u_{yy}(x_i, y_j) \approx \frac{u_{i,j-1} - 2u_{i,j} + u_{i,j+1}}{h^2}.$$

Adding those, we get

$$\frac{u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - 4u_{i,j}}{h^2} = f_{i,j}.$$

So, we have a system of equations for the interior points that we solved for. The Dirichlet boundary conditions were subtracted from the right-hand side. Figure 7 shows the resulting numerical solution.

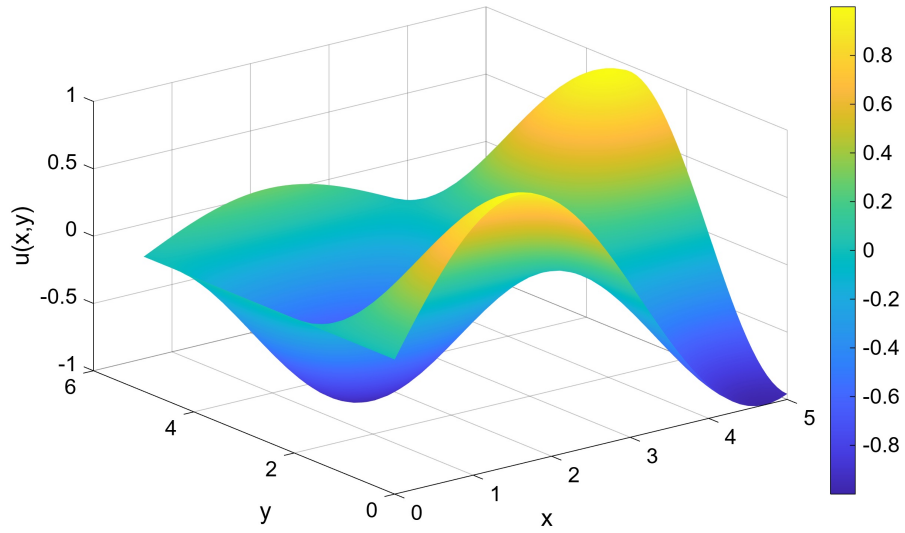


Figure 7: Numerical solution to the two-dimensional Poisson equation using Dirichlet boundary conditions.

Relative errors decrease as the grid granularity increases. The infinity-norm slope is approximately -2.043 and the 2-norm slope is approximately -2.023 , which are both close to the expected slope of -2 . Therefore, the finite difference approximation demonstrates second-order convergence.

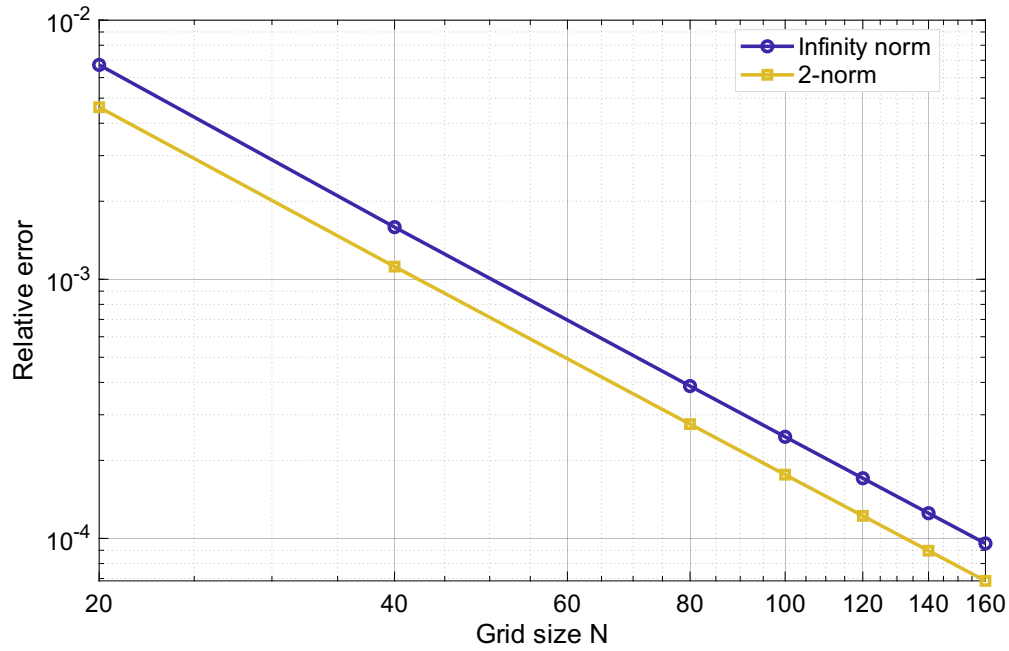


Figure 8: Relative error convergence for the finite difference solution. The $O(N^{-2})$ reference line indicates second-order convergence.

2. Modify your code to apply Neumann conditions on all boundaries. This won't work. Why not—both in terms of the linear algebra and in terms of the actual PDE we want to solve? Suggest a solution to the issue.

Solution:

With Neumann boundary conditions, the solution is not unique. If $u(x, y)$ is a solution, then for any constant C ,

$$\nabla^2(u + C) = \nabla^2 u = f,$$

Since the derivative of a constant is zero, we can say

$$\frac{\partial(u + C)}{\partial n} = \frac{\partial u}{\partial n}.$$

Thus, both u and $u + C$ for any value of C are solutions to the PDE with Neumann boundary conditions.

In terms of the linear algebra, we can write the problem as

$$A\mathbf{u} = \mathbf{b},$$

where A is our linear system of equations, $\mathbf{b}$ is the vector of solutions, and $\mathbf{u}$ is our flattened vector of unknowns. The solution to this equation is not unique; any constant vector is in the null space of A and the system does not have a unique solution.

One solution to this problem could be to specify the value of u at one point, effectively "pinning" one solution.